

Article

Divergence Regarding ESG. A Bibliometric Analysis

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Abstract: *ESG criteria serve as a cornerstone for sustainable business practices, but the growing emphasis on sustainability has led to a surge in ESG rating providers, introducing significant challenges. ESG ratings are inherently complex and often differ across agencies, due to a lack of standardized parameters, risks of green-washing, and the perceived trade-off between profitability and sustainability. Inconsistent methodologies and fragmented regulatory frameworks hinder meaningful comparisons, resulting in limited confidence among investors and companies regarding the reliability of these ratings. Nevertheless, ESG advocates view these ratings as drivers of long-term financial and societal benefits. Consequently, researchers are increasingly focused on understanding the extent of ESG rating divergence, identifying its underlying causes, evaluating its impact on company performance, and exploring strategies to mitigate the negative effects of these inconsistencies. This paper provides an overview of the topic of “ESG ratings divergence,” a subject that has seen exponential growth in the past three years. Using a bibliometric approach, it examines the academic literature published and indexed in the Scopus database with the help of VOSviewer.*

Keywords: *Bibliometric analysis, ESG, ESG rating divergence, ESG score divergence, discrepancies*

1. Introduction

Although the concept of sustainability and sustainable development has been in use for over 50 years, its application in the financial world is more recent. Since 2004, when the term ESG was officially introduced by the UN Global Compact Initiative with the publication of the "Who Cares Wins" report, ESG has evolved into a proactive movement. By the late 2010s and into the 2020s, it developed into a comprehensive framework that incorporates environmental and social impact considerations, alongside strategies to strengthen governance structures and improve stakeholder well-being.

As academics and finance professionals increasingly focused on including sustainability elements in non-financial reporting, the need to rank ESG elements emerged, leading to an explosive growth of ESG rating providers. However, alongside this development, issues related to the comparability of ratings provided by different agencies emerged, driving confusion and lack of trust among investors.

For the past decades, sustainability reporting has largely been voluntary, with disclosure requirements varying significantly by country and region, and divergent regulations introduced to encourage corporate ESG information sharing. However, it is becoming increasingly mandatory in many countries. In response to the rising demand for ESG information from investors and the limited supply from companies, several countries have implemented mandatory ESG disclosure regulations, requiring firms to disclose relevant ESG information either within traditional financial reports or through independent reports (Krueger et al. 2021).

In response to the rising demand for reliable ESG data, the sustainability rating market has grown noticeably, driving to the proliferation of environmental, social, and governance (ESG) metrics and

rating providers, including raw sustainability data, ratings and rankings, indices and benchmarks, consulting services, and reporting practices (Capizzi et al., 2021). Currently some of the most prominent ESG rating providers dominate the Europe and in the United States ESG rating market: MSCI, Refinitiv, Bloomberg, Sustainalytics (recently purchased by Morningstar), S&P Global, VigeoEiris, and Inrate (Capizzi et al., 2021).

With the rising reliance on ESG ratings by market participants, an increasing body of academic research has explored their association with various key variables, such as stock market performance, accounting outcomes, financial constraints, and governance attributes.

Thomas et al. (2024) performed a strategic mapping on a sample of 441 documents retrieved from the Web of Science database, identifying the obsolete, over-researched and promising areas of ESG research. The findings shows that CSR, socially responsible investments and ESG-firm performance themes are over-researched areas, while ESG related portfolio construction, green innovations and investments, controversies surrounding ESG practices, information asymmetry, score divergence, green-washing concerns, and the emergence of AI-enabled universal rating mechanisms seems to be promising areas of ESG research.

2. ESG rates divergence

In the last years, significant attention has been focused on divergence of ratings issued by different ESG rating providers for the same company. If initially, research focused on the correlations between financial performance and sustainability performance, more recently, attention has shifted to the causes behind the divergence of ratings, how these divergences affect returns, and ways to mitigate the negative aspects of these inconsistencies.

Apart from the large number of rating providers, Rating agencies have developed their own assessment methodology to evaluate ESG engagement (Billio et al., 2020), that resulting in divergences among rating agencies in assigning ratings to individual firms (Christensen et al., 2021) .

The 2018 technical report by the American Council for Capital Formation, *"Ratings that Don't Rate – The Subjective View of ESG Rating Agencies"*, highlights significant disparities in ESG ratings among major agencies, attributed to factors such as disclosure limitations, lack of standardization, company size bias, geographic bias, industry sector bias, inconsistencies between agencies, and a failure to identify risks (Doyle, 2018).

Berg et al. (2022) examine the divergence in ESG ratings across six prominent agencies, finding correlations between ESG ratings ranging from 0.38 to 0.71. Abhayawansa and Tyagi (2021) investigate the causes behind the differences in ratings and rankings produced by various agencies, finding that these divergences stem from variations in the definitions of ESG constructs and differences in methodologies used by the rating agencies.

Christensen et al. (2021) examine whether a firm's ESG disclosure helps address the disparity in ratings, finding that ESG disclosure generally amplifies, rather than resolves, the disagreement among ESG ratings.

Billio et al. (2020), examine the ESG rating criteria used by leading agencies and uncover a significant lack of standardization in defining the characteristics, attributes, and standards for the environmental, social, and governance components. This inconsistency leads to divergent assessments, with agencies often presenting conflicting views on the same companies and showing low overall agreement. Furthermore, disagreements in ESG scores diminish the influence of investor preferences on asset prices, so even when ratings align, they fail to have a meaningful impact on financial performance.

3. Data and Methodology

For this paper, a bibliometric analysis of journal articles indexed in Scopus Elsevier database was conducted, using VOSviewer, (vosviewer.com) version 1.6.19, updated on January 2023, for constructing and visualization bibliometric networks, illustrating connections among publications, authors, research organizations, countries, or keywords.

The research was conducted by performing a query within the database, targeting the Article Title, Abstract, and Keywords fields. Boolean logic (AND/OR) was applied to extract the relevant literature corpus, focusing on the reference strings “ESG” and “divergence.”, but also including similar concept: ESG, Economic, Social and Government, ESG rates, ESG scores, ESG disclosures, and divergence, discrepancies, rates divergence, discrepancy, published before 2025.

Thereby, the search consisted in the following (advanced) query: (TITLE-ABS-KEY (esg) OR TITLE-ABS-KEY (economic, AND social AND gouvernement) OR TITLE-ABS-KEY (esg AND rates) OR TITLE-ABS-KEY (esg AND scores) OR TITLE-ABS-KEY (esg AND disclosure) AND TITLE-ABS-KEY (divergence) OR TITLE-ABS-KEY (discrepancies) OR TITLE-ABS-KEY (rating AND divergence) OR TITLE-ABS-KEY (discrepancy)) AND PUBYEAR > 2008 AND PUBYEAR < 2026 AND (LIMIT-TO (LANGUAGE , "English"))

The first query yielded a total of 131 documents. The search was refined, by applying two filters: Language, and subject area. Only the documents in English language were kept, representing 99% of the total documents, excluding 1 document in Russian language and 3 documents in German language.

Since the selected documents fell into multiple disciplinary areas, and their number for each domain was reasonably large, a manual check was performed to determine whether each area included articles whose content could be associated with the theme of the present research, namely the divergence of ESG ratings. Following this verification, certain domains were excluded: Chemistry, Engineering, Medicine, Nursing, Neuroscience, Health Professions. The remaining 103 documents were processed for bibliometric analysis and systematic literature review.

An initial analysis of the search results was conducted using SCOPUS's *Analyze Search Results* tool, focusing on documents categorized by year, publication sources, affiliations, authors, countries/territories, subject areas, and funding sponsors. A more comprehensive bibliometric analysis was then conducted, using VOSviewer, a widely recognized tool for bibliometric research.

4. Primary Analysis of Search Results

A primary analysis of the search results was made, according To Scopus “anlyze results” tool, investigating documents distribution by year, tipe, affiliation, country, funding sponsor and subject area.

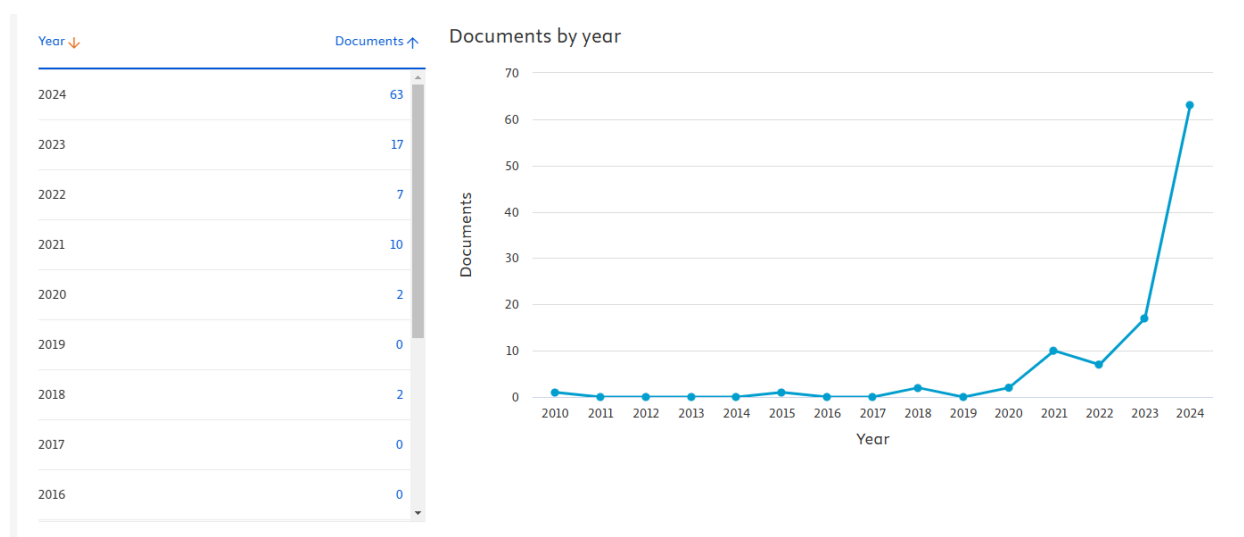


Figure 1. Documents distribution by year 2010-2024. Source: Scopus.com

Figure 1 presents a summary of documents published annually, between 2010 and 2024, highlighting a notable increase in publications addressing the divergence of ESG ratings starting in 2020. This growth has shown an exponential trend over the past three years, considering that 90% of the analysed

documents were published since 2022, reflecting a significant and growing interest among researchers in addressing divergences and discrepancies related to ESG ratings.

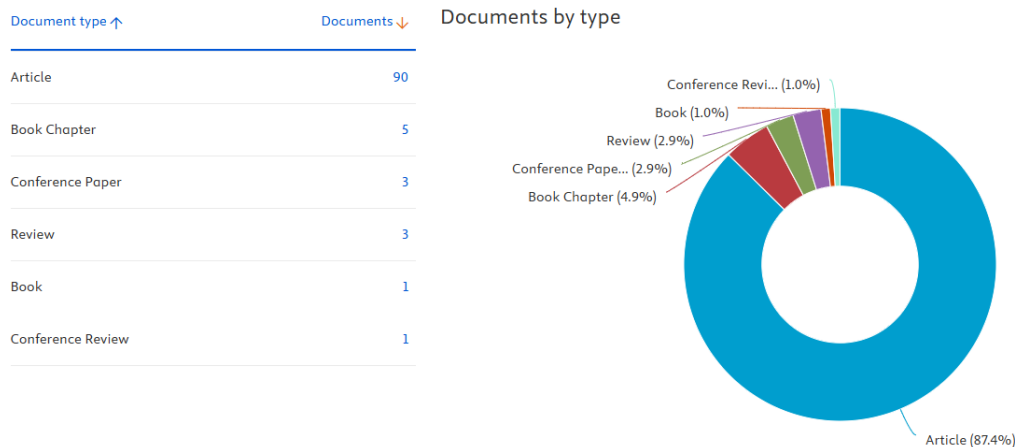


Figure 2. Documents distribution by type. Source: Scopus.com

In terms of document types, 90 records (87% of the total) belong to the Article category, followed by 5 classified as Book Chapters and 3 as Reviews (figure 2). The classification is completed by 1 Conference Papers, 1 Book and 1 Conference Review.

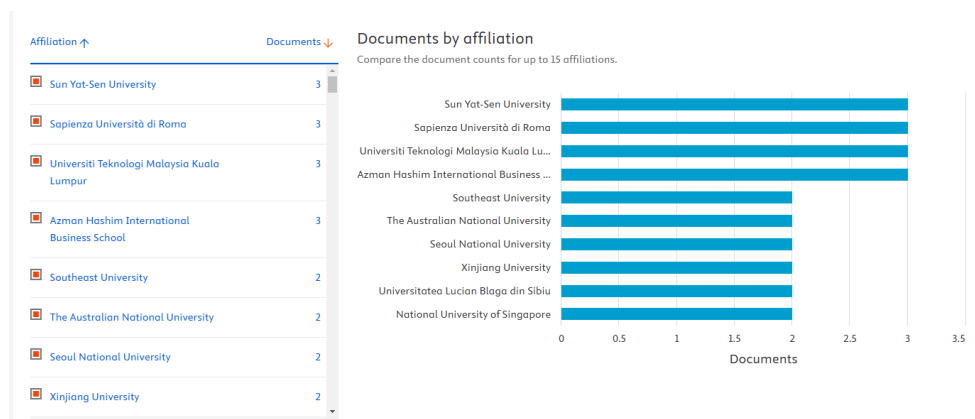


Figure 3. Documents distribution by affiliation. Source: Scopus.com

In terms of document distribution by affiliation, three documents are associated with each of the following institutions (figure 3): Sun Yat-Sen University, Sapienza Università di Roma, Universiti Teknologi Malaysia Kuala Lumpur, Azman Hashim International Business School. Two documents each are affiliated with the following institutions: Southeast University, The Australian National University, Seoul National University, Xinjiang University, Universitatea Lucian Blaga din Sibiu, National University of Singapore, all the aforementioned institutions represent the top 10 institutions in terms of affiliation.

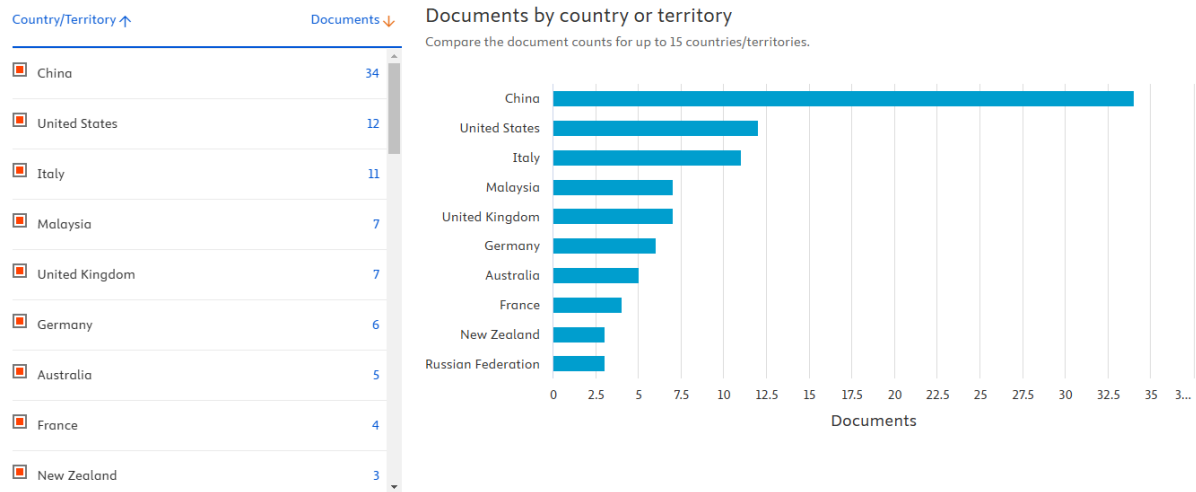


Figure 4. Documents distribution by country Source: Scopus.com

Regarding the distribution of publications by country, the top ten countries where researchers have explored issues related to ESG rating divergences are as follows (figure 4): China (33% of all publications), United States (12% of all publications), Italy (11% of all publications), United Kingdom and Malaysia (7% of all publications each), Australia (5% of all publications), Germany (6% of all publications each), France (4% of all publications) and New Zealand (3% of all publications). Moreover, Romania has recorded 2 publications in this regard.

It should be highlighted that in China can be found most articles related to the divergence of ESG ratings, along with the USA, representing 45% of the total analysed documents. China's top placement can also be attributed to funding sponsors, since the primary analysis of the search results, identifies the National Natural Science Foundation of China (10 documents) Ministry of Science and Technology of the People's Republic of China (7 documents), China Postdoctoral Science Foundation (3 documents) and Ministry of Education of the People's Republic of China (7 documents) as the top five funding sponsors (Figure 5).

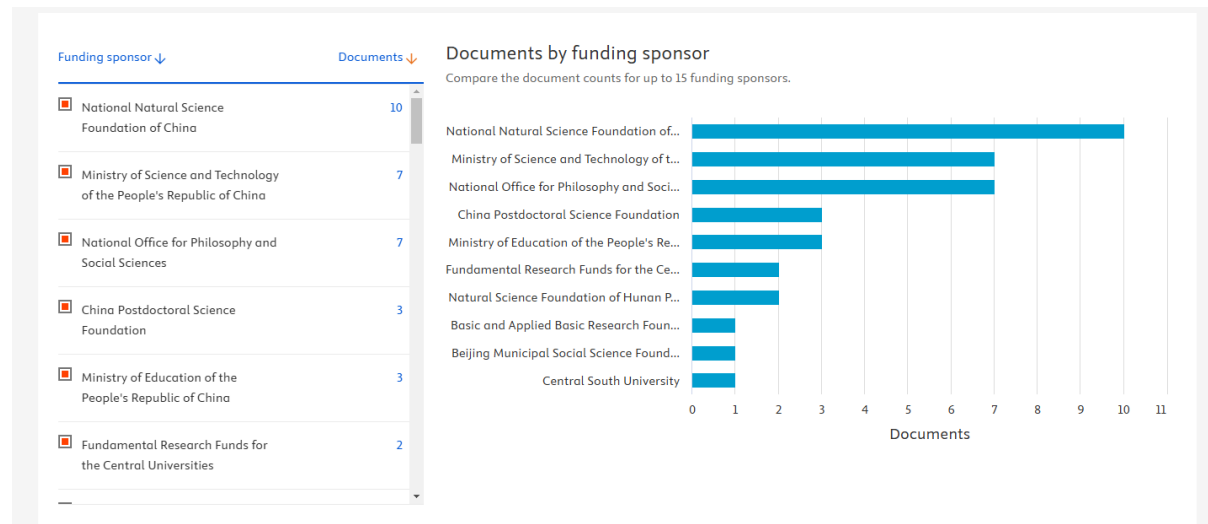


Figure 5. Documents distribution by funding sponsor. Source: Scopus.com

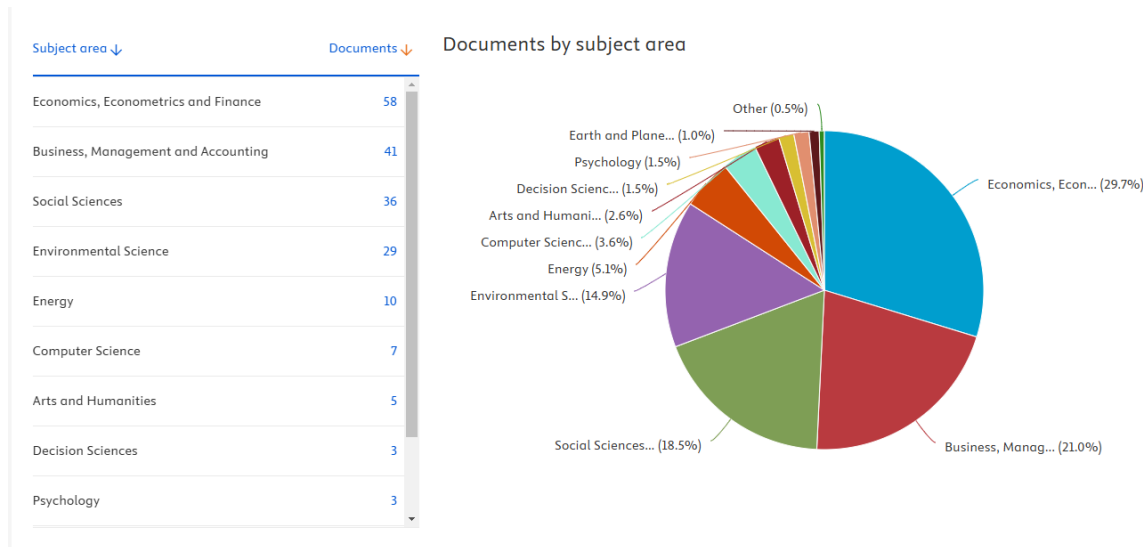


Figure 6. Documents distribution by subject area. Source: Scopus.com

As regards the Research Areas (Figure 6), Economics, Econometrics and Finance (58 records, representing 30%) and Business, Management and Accounting (41 records, representing 21%) are the most frequent areas, together representing almost 51 % of the analysed documents. Social Sciences (36 records, representing 18.5%) Environmental Sciences (29 records, representing 14.9%), Energy (10 records, representing 5,1 %), Computer Science (3,6%), Arts and Humanities (2,6%), Decision Sciences (1,5%), Psychology (1,5%), represent together 44%. The remaining 3%, belong to Earth and Planetary Sciences and Multidisciplinary.

Analysing the trends in the growth of publications and citations related to ESG divergences over time, using Scopus Citation Overview, shown an exponential trend over the past four years.

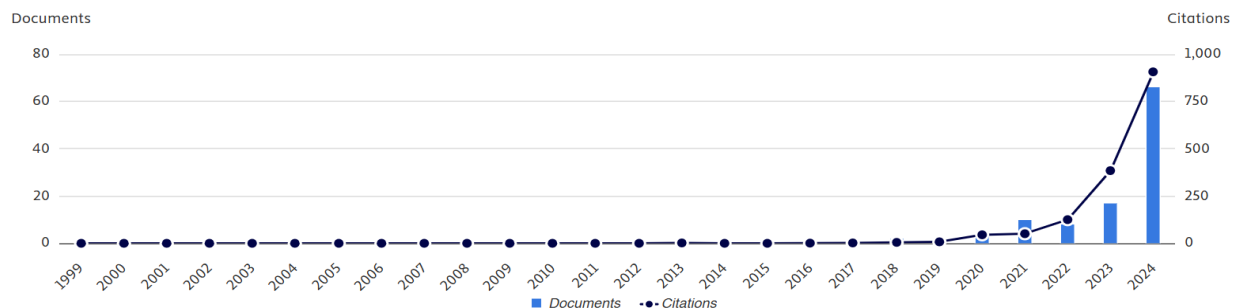


Figure 7. Trends in the growth of publications and citations related to ESG divergences over time. Source: Scopus Citation Overview

Starting from 1999, 64 documents have been cited 1378 times. However, as can be seen in the figure 7, in recent years there has been an exponential increase in the number of citations, with the number of documents following the same trend. In 2024, 857 citations were recorded (62% of citations registered during the analysed period).

Table 1. Top most cited articles related to ESG rating divergence topic

Document Title	Author	Year	Cited before 2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
Aggregate Confusion: The Divergence	Berg; Kölbel; Rigobon	2022	23	0	0	0	0	0	20	151	391	562

of ESG Ratings*												
CSR and idiosyncratic risk: Evidence from ESG information disclosure	Feng, Qin, Liu, Wu	2022	5	0	0	0	0	0	11	57	48	116
Sustainable investing: The black box of environmental, social, and governance	Abhayawansa, Tyagi	2021	0	0	0	0	0	2	13	18	29	62
A survey on ESG: investors, institutions and firms	P. Raghavendra Rao, Ting Yu	2024	5	0	0	0	0	0	0	9	45	54
Do female CEOs matter for ESG scores?	Aabo, Giorigi	2023	0	0	0	0	0	0	0	8	48	56
Management by objectives and corporate social responsibility disclosure		2015	1	2	5	5	9	4	4	6	8	43
Disaggregating confusion? The EU Taxonomy and its relation to ESG rating		2022	4	0	0	0	0	0	2	12	23	37
Esg rating—necessity for the investor or the company?		2021	0	0	0	0	0	0	8	11	21	40
Fundamental ratios as predictors of ESG scores: a machine learning approach		2021	1	0	0	0	0	0	3	12	17	32
Corporate ESG rating divergence and excess stock returns		2024	6	0	0	0	0	0	0	0	24	24
Quantitative ESG disclosure and divergence of ESG ratings		2022	1	0	0	0	0	0	0	7	21	28
The Divergence Of Esg Ratings: An Analysis Of Italian Listed Companies		2021	0	0	0	0	0	0	0	6	19	25
Doing well by doing good: A comparative		2018	0	0	0	2	1	1	2	7	10	23

analysis of ESG standards												
ESG rating divergence and corporate green innovation	2024	3	0	0	0	0	0	0	0	0	13	13
Non-Financial Reporting & Corporate Governance	2020	0	0	0	0	0	1	4	6	5	16	

Source: Citation overview.Scopus.com

Top five most cited articles, with over 50 citations each, together summing up over 60% of the total citations, focus on research the causes that led to the divergence of ESG rates (table 1).

The most cited work (566 citations) belongs to Berg, Kölbel and Rigobon. In the article, “*Aggregate Confusion: The Divergence of ESG Ratings*” published in 2022, the authors (Berg et al., 2022) investigate the divergence of environmental, social, and governance (ESG) ratings based on data from six prominent ESG rating agencies (Kinder, Lydenberg, and Domini (KLD), Sustainalytics, Moody’s ESG (Vigeo-Eiris), S&P Global (RobecoSAM), Refinitiv (Asset4), and MSCI), decomposing the divergence into contributions of scope, measurement, and weight. Findings show that measurement contributes 56% of the divergence, scope 38%, and weight 6%.

The second most cited article (121 citations in Scopus), “*CSR And Idiosyncratic Risk: Evidence From ESG Information Disclosure*”, Feng et al. (2022) conducted an empirical research to find the impact of ESG information disclosure on firms’ idiosyncratic risk in the Chinese stock market where retail investors play a dominant role. Findings indicate that ESG information disclosures improve the informational efficiency of the stock market while reducing a firm's idiosyncratic risk, this impact being more pronounced among companies subject to mandatory CSR disclosure requirements. Thus, providing additional non-financial information, CSR disclosures help mitigate opinion divergence among investors.

Abhayawansa and Tyagi (2021) explores the of discrepancies in ESG ratings and rankings across different agencies., finding that these divergences arise from variations in the definitions of ESG constructs, methodological approaches, and a lack of transparency regarding data sources, weightings, and evaluation processes. These inconsistencies hinder the accurate assessment of companies' true ESG performance, complicating portfolio selection and investment decisions.

In their survey, Rau and Yu (2024) review academic literature on ESG/CSR from the perspectives of investors, institutions, and firms. They examine challenges in ESG measurement, including issues with data quality and inconsistencies among metrics. From the investors' perspective, the study explores the types of investors involved in ESG and the role of institutional investors. From the firms' perspective, it analyses the motivations for adopting ESG practices and their impact on financing, disclosure, reporting, and performance. Rau and Yu summarize ESG measurement methods, the evolution of major ESG rating agencies and the key factors contributing to divergences in ESG ratings. They also highlight inconsistent empirical findings regarding the relationship between firm-level CSR and firm performance, while exploring the broader implications of ESG/CSR adoption.

Abao and Giorici (2023) tuck an approach to ESG ratings through the lens of gender diversity, investigating the relationship between CEO gender and corporate social responsibility. They also questioned the inconsistencies among data providers in measuring ESG scores, raising concerns about the quality of individual ESG metrics and highlighting the significant variations in ESG scores across different providers. If no association between CEO gender and corporate social responsibility was found when analysis was based on ESG data from Refinitiv, such a correlation was found when data from Bloomberg was used. The discrepancy in results stem primarily from the Environmental and Social pillars. These differences cannot be attributed to non-identical samples, since the analysis was based on companies for which both Bloomberg and Refinitive provided data. Thus, the discrepancies are due to methodological differences between the two ESG data providers rather than variations in firm coverage.

5. Bibliometric Analysis Using VOSviewer

Secondly, the data obtained from Scopus was exported in CSV format and imported into VOSviewer software, to identify research clusters and generate network-based . Two types of analysis were run: citation analysis and keywords co-occurrence analysis (using all author's keywords)

5.1. Citation analysis

Related to the previous statements about the citations, the citation analysis with “authors” as unit of analysis was performed, considering minimum one documents of an author and minimum five citation of an author. 81 authors out of 287 meet the thresholds, lading to 2 clusters and 88 links.

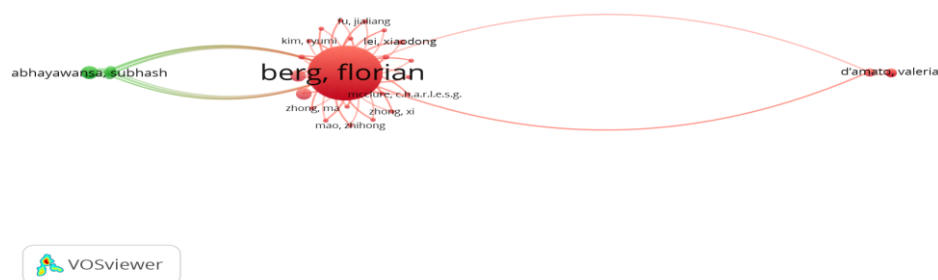


Figure 8. Citation map of authors. Source VOSviewer

The most cited authors, considering the citations are: Berg, Kölbel and Rigobon (585 citations and a total link strength of 26), the three being co-authors of the paper “*Aggregate Confusion: The Divergence of ESG Ratings*”. They are followed by Feng, Qin, Liu, Wu (121 citations) and by Abhayawansa and Tyagi (121 citations). It can be observed that the authors with the highest number of citations are also the authors of the most cited articles, with the hierarchy of the most cited authors mirroring the hierarchy of their most cited works (figure 8).

The fact that 81 authors represent only 28% of the total number of authors should be interpreted with caution, both because the articles were published recently, primarily in the last three years, and due to the number of co-authors per paper. For instance, among the 81 authors, 2 or 3 are often co-authors of the same work.

Table 2. Top most cited authors, based on citations number, related to ESG rating divergence topic

Author	Citations	Total Link Strength	Author	Citations	Total Link Strength
Berg, Florian	585	26	Ge, Chen	30	5
Kölbel, Julian F.	585	26	Jiao, Shuaipeng	30	5
Rigobon, Roberto	585	26	Sun, Guanglin	30	5
He, Feng	121	0	Wang, Haijun	30	5
Liu, Yuanyuan	121	0	Liu, Min	29	0
Qin, Shuqi	121	0	Capizzi, Vincenzo	25	0
Wu, Ji (George)	121	0	Gioia, Eleonora	25	0
Abhayawansa, Subhash	62	4	Giudici, Giancarlo	25	0
Tyagi, Shailesh	62	4	Tenca, Francesca	25	0

Rau, P. Raghavendra	59	3	Barman, Emily	23	0
Yu, Ting	59	3	Harper Ho, Virginia	16	0
Aabo, Tom	56	0	Lei, Xiaodong	16	3
Giorici, Iasmina Cristina	56	0	Yu, Jianglong	16	3
Leopizzi, Rossella	44	0	Zhou, Jian	16	3
Mio, Chiara	44	0	Garcia-Bernabeu, Ana	15	0
Venturelli, Andrea	44	0	Pla-Santamaria, David	15	0
Dumrose, Maurice	41	0	Reig-Mullor, Javier	15	0
Eckert, Julia	41	0	Vercher-Ferrandiz, Marisa	15	0
Rink, Sebastian	41	0	Feng, Xuan	13	0
Lāce, Nataļja	40	0	Cardillo, Giovanni	12	0
Zumente, Ilze	40	0	Harasheh, Murad	12	0
D'amato, Valeria	33	1	Horn, Matthias	11	0
D'ecclesia, Rita	33	1	Kim, Ryumi	10	3
Levantesi, Susanna	33	5	Koo, Bonha	10	3

Source: VOSviewer

The citation analysis with "source" as unit of analysis was performed considering minimum 1 document of a source and minimum five citation of a source (figure 9). Of the 75 sources, 27 meet the thresholds, generating two clusters, with 14 links and 20 total link strength. In cluster 1 (red) "Review of finance" was cited most times (585), breaking away significantly from "Journal of Wealth Management" in Cluster 2 (green), who recorded only 62 citations. The first 10 sources ranked from the perspective of the number of citations are Review of Finance (585 citations), Finance Research Letters (182 citations), Journal Of Wealth Management (62 citations), China Finance Review International (59 citations), Global Finance Journal (57 citations), Sustainability (Switzerland) (46 citations), Accounting, Auditing And Accountability Journal (44 citations), Decisions In Economics And Finance (33 citations), Energy Economics (30 citations), Frontiers In Psychology (29 citations).

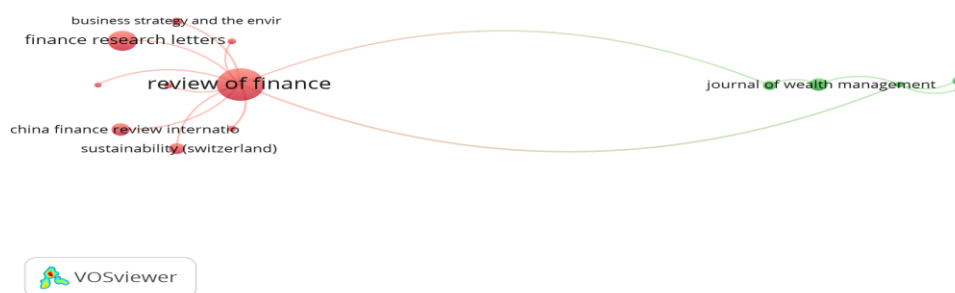


Figure 9. Citation map of sources. Source VOSviewer

5.2. Co-occurrence of author's keywords

The analysis of author keyword co-occurrence was prioritized over the co-occurrence of all keywords, considering that it better reflect the authors' perspectives on the researched topics (Ogrean and Herciu,

2021). The map was generated using VOSviewer with a keyword occurrence threshold set to 5 (figure 10). Following the keyword co-occurrence analysis, in a full counting mode, of all 337 authors' keywords, 49 terms were identified as meeting the threshold of appearing in at least two documents. Six distinct clusters have emerged, with 673 links and 841 total link strength. (Figure 8 and Table 2, organized around the following keywords: *sustainability*, *environmental*, *ESG ratings*, *ESG disclosure*, *ESG rating divergence*, *CSR*, *ESG investing*. ESG and sustainability are the most used keywords, strongly connected with the rest of the keywords, especially ESG rating divergence, sustainable reporting, green-washing, sustainable finance and firm performance.

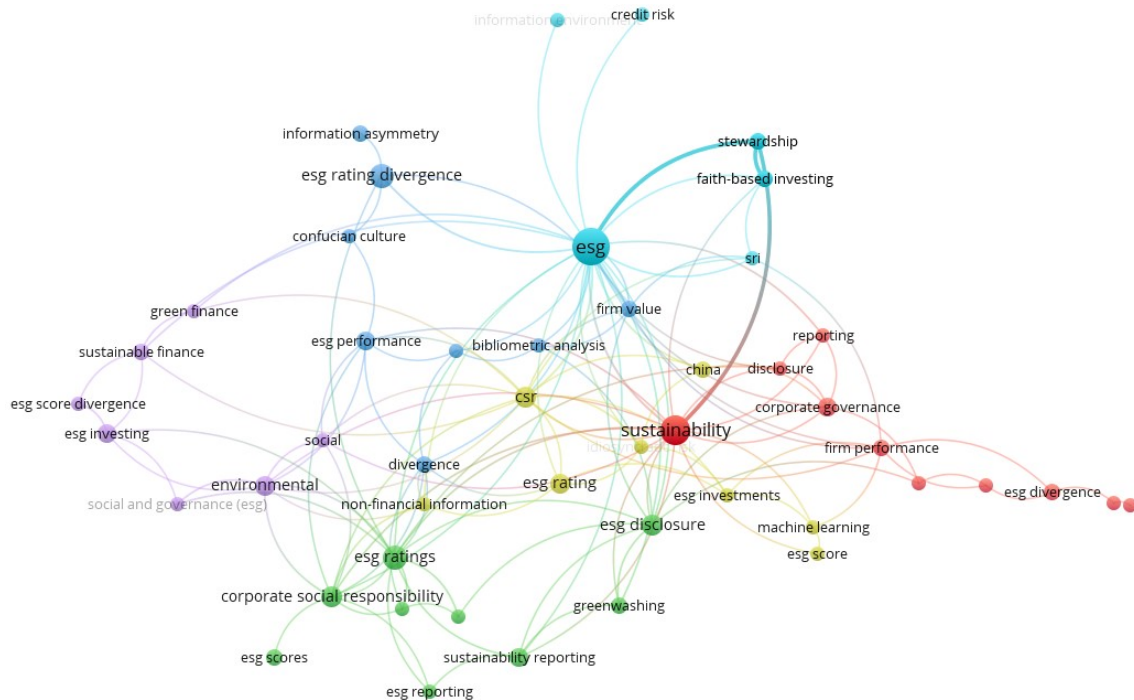


Figure 10. Author keyword co-occurrence map. Source: VOSviewer

Table 3 shows the most used keywords by the authors in ESG rates divergence articles and the keywords with the most powerful total link strength for each cluster.

Table 3. Author keywords co-occurrence clusters

Cluster	All Keywords	Keywords with the most powerful total link strength
Cluster 1 (Red)-10 items,	corporate governance, disclosure, environmental regulation, esg divergence, firm performance, firm value, industry competition, media attention, reporting, stock price crash risk, sustainability	<i>sustainability</i> 14 occurrence, 26 total link strength
Cluster 2 (green) - 9 items	corporate social responsibility, corporate sustainability, esg disclosure, esg ratings, esg reporting, esg score, greenwashing, rating divergence, sustainability reporting	<i>ESG ratings</i> - 9 occurrence, 20 total link strength <i>ESG disclosure</i> - 6 occurrence, 11 total link strength
Cluster 3 (blue) -8 items	bibliometric analysis, confucian culture, divergence, environmental social governance, esg performance, esg rating	<i>ESG rating divergence</i> - 9 occurrence, 8 total link strength

	divergence, firm value, information asymmetry	
Cluster 4 (Olivegreen)- 8 items	China, CSR, esg investments, esg ratings, esg score, idiosyncratic risk, machine learning, non-financial information	CSR - 6 occurrence, 20 total link strength ESG rating - 5 occurrence, 6 total link strength
Cluster 5 (Mauve)- 7 items	Environmental, esg investing, esg score, divergence, green finance, greenwashing Social, social and governance (esg), sustainable finance	Environmental - 5 occurrence, 13 total link strength ESG investing - 4 occurrence, 6 total link strength
Cluster 6 (Turquoise)- 6 items	credit risk, ESG, faith-based investing, information environment, SRI, stewardship	ESG - 29 occurrence, 50 total link strength

In addition to *ESG rate divergences* the most influential keywords in terms of total link strength and occurrences, are: *ESG* (with 29 occurrences and a total link strength of 50), *sustainability* (with 14 occurrences and a total link strength of 26), *ESG ratings* (with 9 occurrences and a total link strength of 20) and *CSR* (with 5 occurrences and a total link strength of 20).

The keywords related to divergence occurs as “*esg rating divergence*” (with 9 occurrences and a total link strength of 8), “*divergence*”, (with 3 occurrences and a total link strength of 8) “*green washing*”, (with 3 occurrences and a total link strength of 6), “*esg rating divergence*” (with 9 occurrences and a total link strength of 31), “*esg divergence*”, (with 3 occurrences and a total link strength of 3) “*esg score divergence*” (with 2 occurrences and a total link strength of 3), “*rating divergence*” (with 2 occurrences and a total link strength of 4), cumulating 5% of all occurrences and 4% of total link strength.

6. Conclusions

While CSR, socially responsible investments and ESG-firm performance themes are over-researched areas, ESG related portfolio construction, green innovations and investments, controversies surrounding ESG practices, information asymmetry, score divergence, green-washing concerns, and the emergence of AI-enabled universal rating mechanisms seems to be promising areas of ESG research. Researchers' concerns about ESG rating divergence have experienced exponential growth over the past three years, both in the number of articles and in the number of citations.

The study offers a comprehensive bibliometric analysis of ESG rating divergences, utilizing Scopus-indexed literature and VOSviewer to identify key trends, clusters, and influential contributions in the field. The research was conducted by performing a query within the Scopus database, targeting the Article Title, Abstract, and Keywords fields, focusing on the reference strings “ESG” and “divergence.”, but also including similar concept. The first query of 131 documents was refined, by applying language and subject area filters. The remaining 103 documents were processed for bibliometric analysis and systematic literature review.

A primary analysis of the search results was made, according to Scopus “analyze results” tool, investigating documents distribution by year, tipe, affiliation, country, funding sponsor and subject area.

A significant finding is the exponential increase in publications since 2020, considering that 90% of the analysed documents were published since 2022, reflecting a significant and growing interest among researchers in addressing divergences and discrepancies related to ESG ratings. 87% of the total documents belong to the Article category, Economics, Econometrics and Finance and Business, Management and Accounting being the most frequent areas. Geographically, China leads in research output, supported by significant funding, followed by the USA and European countries.

Analysing the trends in the growth of publications and citations related to ESG divergences over time, using Scopus Citation Overview, an exponential increase in over the past four years was noted.

This analysis was complemented by a bibliometric analysis using VOSviewer, with two types of analysis being conducted: citation analysis and keywords co-occurrence analysis.

The most cited document “*Aggregate Confusion: The Divergence of ESG Ratings*” belongs to Berg, Kölbel and Rigobon, who are also the most cited authors, reveals that measurement issues contribute the most to rating divergence (56%), followed by scope (38%) and weighting (6%). The top 5 most cited articles also include: “*Aggregate Confusion: The Divergence of ESG Ratings**”, “*CSR and idiosyncratic risk: Evidence from ESG information disclosure*”, “*Sustainable investing: The black box of environmental, social, and governance*”, “*A survey on ESG: investors, institutions and firms*”, “*Do female CEOs matter for ESG scores?*” the authors with the most citations being the authors of these articles.

The most cited document highlights that ESG ratings often lack standardization, with divergences arising from methodological, scope, and weighting differences among major rating agencies. These inconsistencies complicate corporate assessments and sustainable investment decisions, reducing investor confidence.

The analysis of author keywords led to six distinct clusters, with 673 links and 841 total link strength, organized around the following keywords: *sustainability, environmental, ESG ratings, ESG disclosure, ESG rating divergence, CSR, ESG investing*. ESG and sustainability are the most used keywords, strongly connected with the rest of the keywords, especially ESG rating divergence, sustainable reporting, green-washing, sustainable finance and firm performance.

In addition to *ESG rate divergences* the most influential keywords in terms of total link strength and occurrences, are: *ESG, sustainability, ESG ratings* and *CSR*. The keywords related to divergence occurs as “*esg rating divergence*”, “*divergence*”, “*green washing*”, “*esg rating divergence*”, “*esg divergence*”, “*esg score divergence*”, “*rating divergence*”, cumulating 5% of all occurrences and 4% of total link strength.

The limitations of the research is due to the relatively small number of articles, as researchers' interest in ESG rating divergence has primarily emerged over the past three years. To gain a deeper understanding of the specialized literature, the analysis should also be extended to include databases provided by WoS and other database supported by VOSviewer.

Conflicts of Interest

The author declares that there are no conflicts of interest regarding the publication of this paper.

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